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SABR MODELS FOR SYNTHETIC FX RATES

PATRICK S. HAGAN*, ANDREW S. LESNIEWSKI†, G. E. SKOUFIS‡, AND DIANA E. WOODWARD§

Abstract. Let X_{ij} be the forward FX rate for currency i in terms of currency j . Suppose that the volatility smiles for X_{ik} and X_{jk} are modeled with the log normal ($\beta = 1$) SABR model. We show that the cross FX rate $X_{ij} \equiv X_{ik}/X_{jk}$ is then governed by the log normal SABR model, and find explicit formulas for its SABR parameters α, ρ, ν in terms of the SABR parameters for X_{ik} and X_{jk} . These results are not exact, but they are accurate through $O(\varepsilon^2)$, the same accuracy as the original SABR implied volatility formulas. We also extend this analysis to self-consistent models for several currencies, where each currency pair can be driven by multiple factors.

Consider foreign exchange rates for currencies i and j in terms of a common base currency, typically USD. Often there are deep and liquid option markets for these FX rates, but only weak or non-existent option markets for the cross FX rates of currency i in terms of currency j or currency j in terms of i . One naturally wonders whether we can determine the option prices and volatility smiles for these cross FX rates from the market prices and smiles of the currency i /base and currency j /base options. This would allow us to inject liquidity into the market for currency i /currency j options, hedging our risks in the more liquid markets. I.e., we would be able to synthetically re-create options on the cross FX rate by dynamically hedging in the currency i /base and currency j /base markets. Here we show that if the log normal ($\beta = 1$) SABR model is used to model the currency i /base and currency j /base options, then the option prices and volatility smiles of the cross FX rates are also governed by the log normal SABR model, and we obtain explicit formulas for the SABR parameters for these options in terms of the SABR parameters of the currency i /base and currency j /base options. These results are not exact, but they are accurate through $O(\varepsilon^2)$, the same accuracy as the original SABR implied volatility formulas.

For some major currencies there are well-developed option markets for the currency i /currency j cross rates, as well as for the currency i /base and currency j /base rates. In this case our results determine if the volatility smile of the cross FX rate is consistent with the volatilities of the component FX rates. If not, we can synthetically re-create the cross FX options by dynamically hedging with options on the component FX rates, and use these synthetic options to arbitrage the market.

We also create self-consistent models for an arbitrary number of currency pairs, models where each currency pair can be driven by multiple factors. We extend our analysis to these models, obtaining the effective SABR model and effective SABR parameters for each currency pair.

1. Cross FX rates. Let us choose a specific exercise date T_{ex} , and let T_{set} be the corresponding settlement date. Let currency 0 be the base currency, and define the forward FX rates for this settlement date as

$$(1.1) \quad \tilde{X}_{i0} = \frac{\text{ccy } i}{\text{ccy } 0} = \text{forward price of 1 unit of ccy } i \text{ in terms of ccy } 0,$$

for currencies $i = 1, 2$. Suppose that the log normal SABR model is used for these FX rates. Then in the forward measure¹,

$$(1.2a) \quad d\tilde{X}_{i0} = \varepsilon \tilde{A}_i \tilde{X}_{i0} dW_i,$$

$$(1.2b) \quad d\tilde{A}_i = \varepsilon \nu_i \tilde{A}_i dZ_i,$$

$$(1.2c) \quad dW_i dZ_i = \rho_{ii} dT,$$

*Gorilla Science, PatHagan@GorillaSci.Com.

†Dept of Mathematics; Baruch College; One Baruch Way; New York, NY 10010.

‡Santander Corporate & Investment Banking, 2 Triton Square, Regent's Place, London NW1 3AN.

§Gorilla Science, Diana_Woodward@yahoo.com

¹Forward FX rates are Martingales in the denominating currency's forward measure for T_{set} , so there are no drift terms.

for $i = 1, 2$. The SABR parameters $\alpha_1 \equiv \tilde{A}_1(0)$, ρ_{11} , ν_1 and $\alpha_2 \equiv \tilde{A}_2(0)$, ρ_{22} , ν_2 for these models can be obtained by calibrating the theoretical SABR prices to the observed market prices for options on $\tilde{X}_{10}$ and $\tilde{X}_{20}$. This can be done by fitting the observed market smiles to the SABR model's implied volatility formulas, which are reproduced in Appendix C for completeness. Alternatively, one can follow the arbitrage free approach [2],[3], finding the theoretical option prices by using numerical methods to solve the SABR model's effective forward equation, and then fitting these prices to the market's option prices.

Our objective is to determine the market prices for options on the cross FX rate,

$$(1.3) \quad \tilde{X}_{12} = \frac{\tilde{X}_{10}}{\tilde{X}_{20}} = \text{forward price for 1 unit of ccy 1 in terms of ccy 2.}$$

Using Itô's formula to differentiate 1.3 yields

$$(1.4) \quad \begin{aligned} d\tilde{X}_{12} &= \frac{d\tilde{X}_{10}}{\tilde{X}_{20}} - \tilde{X}_{10} \frac{d\tilde{X}_{20}}{\tilde{X}_{20}^2} + \text{drift terms} \\ &= \varepsilon \tilde{X}_{12} \left\{ \tilde{A}_1 dW_1 - \tilde{A}_2 dW_2 \right\} + \text{drift terms.} \end{aligned}$$

The appropriate measure for options on $\tilde{X}_{12}$ is the forward measure for currency 2. In this measure $\tilde{X}_{12}$ is a Martingale, so the drift terms must be identically zero. In this measure, then,

$$(1.5a) \quad d\tilde{X}_{12} = \varepsilon \tilde{X}_{12} \left\{ \tilde{A}_1 dW_1 - \tilde{A}_2 dW_2 \right\},$$

$$(1.5b) \quad d\tilde{A}_1 = \varepsilon \nu_1 \tilde{A}_1 dZ_1,$$

$$(1.5c) \quad d\tilde{A}_2 = \varepsilon \nu_2 \tilde{A}_2 dZ_2.$$

The correlations

$$(1.6a) \quad dW_1 dZ_1 = \rho_{11} dT, \quad dW_2 dZ_2 = \rho_{22} dT,$$

are determined by calibrating the individual SABR models for the component FX rates, $\tilde{X}_{10}$ and $\tilde{X}_{20}$. The FX rate to FX rate correlation,

$$(1.6b) \quad dW_1 dW_2 = c dT,$$

is also a critical and easily observable parameter. To complete the model we need the other correlations,

$$(1.6c) \quad dW_1 dZ_2 = \rho_{12} dT, \quad dW_2 dZ_1 = \rho_{21} dT, \quad dZ_1 dZ_2 = r dT.$$

The correlations ρ_{12} , ρ_{21} , and r are not easily observable, and obtaining them by fitting cross FX volatility smiles to market data would overfit the data. However, it has been noted that volatilities of FX rates are usually highly correlated, with FX markets becoming volatile or quiescent in synchrony, or, equivalently, with FX option prices rising and falling together. Accordingly, it has become common practice to ignore any idiosyncracies in the individual FX rate volatilities, and model the FX volatilities as being driven by the same process,

$$(1.7) \quad dZ_1 = dZ_2 \equiv dZ.$$

This is the *single stochastic vol* model. It follows that

$$(1.8) \quad \rho_{12} = \rho_{11}, \quad \rho_{21} = \rho_{22}, \quad r \equiv 1.$$

The advantage of the single stochastic vol model is that the cross FX volatility smile is determined by a single extra parameter, the FX rate to FX rate correlation c . The other parameters are determined by the SABR models for the component FX rates. So if we can fit c to, say, the cross FX rate's at-the-money volatility, the model then determines the rest of the cross FX rate's volatility smile.

1.1. The effective SABR parameters. In Appendix A we analyze a broad class of multifactor log normal SABR models. These models are specialized to the cross FX model of 1.5a-1.5c in Appendix B. There it is shown that at any given exercise date T_{ex} , the cross FX rate has the same probability distribution, and thus the same European option prices, as the effective SABR model

$$\begin{aligned} (1.9a) \quad & d\tilde{X}_{12} = \varepsilon \tilde{A} \tilde{X}_{12} dW, \\ (1.9b) \quad & d\tilde{A} = \varepsilon \nu^{eff} \tilde{A} dZ, \\ (1.9c) \quad & dW dZ = \rho^{eff} dT, \end{aligned}$$

to within $O(\varepsilon^2)$. These prices are encapsulated by the SABR model's explicit formulas for its implied volatilities. For completeness, these formulas are presented in Appendix C.

The values of the SABR parameters $\alpha^{eff} \equiv \tilde{A}(0)$, ρ^{eff} , and ν^{eff} are derived in Appendix B. For the special case of a single stochastic vol, $dZ_1 = dZ_2 \equiv dZ$, where the correlations are given by 1.8, these SABR parameters are²

$$\begin{aligned} (1.10a) \quad & \alpha^{eff} = \Delta(\alpha) e^{\frac{1}{4}\varepsilon^2 \Gamma(\alpha) T_{ex}}, \\ (1.10b) \quad & \rho^{eff} = \bar{\rho} \bar{\nu} / \sqrt{\bar{\nu}^2 + \kappa}, \\ (1.10c) \quad & \nu^{eff} = \sqrt{\bar{\nu}^2 + \kappa}, \end{aligned}$$

where T_{ex} is the time to expiry and³

$$\begin{aligned} (1.11a) \quad & \Delta^2(\alpha) = \alpha_1^2 - 2c\alpha_1\alpha_2 + \alpha_2^2, \\ (1.11b) \quad & \bar{\rho}(\alpha) = \frac{\alpha_1\rho_{11} - \alpha_2\rho_{22}}{\Delta(\alpha)}, \\ (1.11c) \quad & \bar{\nu}(\alpha) = \frac{\alpha_1^2\nu_1 - c\alpha_1\alpha_2(\nu_1 + \nu_2) + \alpha_2^2\nu_2}{\Delta^2}, \\ (1.11d) \quad & \Gamma(\alpha) = (1 - c^2) \frac{\alpha_1^2\alpha_2^2}{\Delta^4} (\nu_1 - \nu_2)^2 - \kappa(\alpha), \end{aligned}$$

with

$$\begin{aligned} (1.11e) \quad & \kappa(\alpha) = \left\{ 3(1 - c^2) \frac{\alpha_1^2\alpha_2^2}{\Delta^4} - 2c \frac{\alpha_1\alpha_2}{\Delta^2} \right\} \bar{\rho}^2 (\nu_1 - \nu_2)^2 \\ & + \frac{\alpha_1\alpha_2}{\Delta^3} \{ (\alpha_1 - c\alpha_2) \rho_{22}\nu_1 + (\alpha_2 - c\alpha_1) \rho_{11}\nu_2 \} \bar{\rho} (\nu_1 - \nu_2). \end{aligned}$$

See B.5a-B.5c and B.7a-B.8c.

Note that $\bar{\rho}$ and $\bar{\nu}$ represent the volatility weighted correlation and vol-of-vol, while Γ and κ arise from the dispersion of vols. It is not uncommon for Γ and κ to be much smaller than $\bar{\nu}^2$. In this case these terms can be omitted, at least for qualitative purposes, so that $\rho^{eff} \simeq \bar{\rho}$, $\nu^{eff} \simeq \bar{\nu}$, and $\Gamma \simeq 0$. Then the effective SABR coefficients are just Δ , $\bar{\rho}$, and $\bar{\nu}$.

1.1.1. General correlation structures. The SABR model 1.9a-1.9c with the effective SABR parameters 1.10a-1.11e are the key results of this paper. By using the “single stochastic vol” assumption

²Note that each expiry date T_{ex} requires a slightly different α^{eff} to obtain the correct European option prices for that expiry.

³Note that these formulas are written in terms of α_1 and α_2 , whereas the formulas in Appendices A and B are written in terms of $a_1 = \lambda_1\alpha_1$ and $a_2 = \lambda_2\alpha_2$ where $\lambda_1 = 1$ and $\lambda_2 = -1$ for cross FX rates.

$dZ_1 = dZ_2 \equiv dZ$, we were able to model the cross FX rate with a single new parameter, the FX rate correlation c . We believe this gives the best balance between accuracy and parsimony of parameters.

Appendix B also derives the effective SABR parameters for the general correlation structure, where

$$(1.12) \quad dW_1 dZ_2 = \rho_{12} dT, \quad dW_1 dZ_2 = \rho_{12} dT, \quad dZ_1 dZ_2 = r dT.$$

To write these effective SABR parameters succinctly, define

$$(1.13a) \quad \rho_{1X} \equiv \frac{\alpha_1 \rho_{11} - \alpha_2 \rho_{21}}{\Delta}, \quad \rho_{2X} \equiv \frac{\alpha_1 \rho_{12} - \alpha_2 \rho_{22}}{\Delta},$$

$$(1.13b) \quad \nu_{1X} \equiv \frac{(a_1 - c\alpha_2) \alpha_1}{\Delta^2} \nu_1, \quad \nu_{2X} \equiv \frac{(\alpha_2 - c\alpha_1) \alpha_2}{\Delta^2} \nu_2,$$

$$(1.13c) \quad \nu_{mix} = \frac{(\alpha_1 \rho_{11} - \alpha_2 \rho_{21}) \nu_1 - (\alpha_1 \rho_{12} - \alpha_2 \rho_{22}) \nu_2}{\Delta}.$$

For this general case, the effective SABR parameters for options with exercise date T_{ex} are

$$(1.14a) \quad \alpha^{eff} = \Delta(\alpha) e^{\frac{1}{4} \varepsilon^2 \Gamma(\alpha) T_{ex}},$$

$$(1.14b) \quad \rho^{eff} = \beta(\alpha) / \sqrt{\gamma(\alpha)},$$

$$(1.14c) \quad \nu^{eff} = \sqrt{\gamma(\alpha)},$$

where

$$(1.15a) \quad \Delta^2(\alpha) = \alpha_1^2 - 2c\alpha_1\alpha_2 + \alpha_2^2.$$

$$(1.15b) \quad \beta(\alpha) = \rho_{1X}\nu_{1X} + \rho_{2X}\nu_{2X},$$

$$(1.15c) \quad \gamma(\alpha) = \nu_{1X}^2 + 2r\nu_{1X}\nu_{2X} + \nu_{2X}^2 + \kappa(\alpha),$$

$$(1.15d) \quad \Gamma(\alpha) = (1 - c^2) \frac{\alpha_1^2 \alpha_2^2}{\Delta^4} (\nu_1^2 - 2r\nu_1\nu_2 + \nu_2^2) - \kappa(\alpha),$$

$$(1.15e) \quad \kappa(\alpha) = \left\{ 3(1 - c^2) \frac{\alpha_1 \alpha_2}{\Delta^2} - 2c \right\} \frac{\alpha_1 \alpha_2}{\Delta^2} \nu_{mix}^2 + \frac{1}{\Delta} \{ \alpha_2 \rho_{21} \nu_{1X} + \alpha_1 \rho_{12} \nu_{2X} \} \nu_{mix}.$$

See B.3a-B.3d and B.5a-B.6e.

1.1.2. Near independent volatilities. In the “single stochastic vol” model, the stochastic vols dZ_1 and dZ_2 are perfectly correlated. Even though this is a good model for real markets, it is natural to look at the opposite limit, where the stochastic volatilities are as independent as possible. Since the two FX rates dW_1 and dW_2 are correlated, and since each FX rate’s volatility dZ_j is correlated with its own FX rate dW_j , it is unreasonable to expect dZ_1 and dZ_2 to be uncorrelated. Instead we assume that

$$(1.16) \quad dZ_j = \rho_{jj} dW_j + \sqrt{1 - \rho_{jj}^2} dZ_j^\perp \quad \text{for all } j = 1, 2,$$

where the Brownian motions $dZ_j^\perp$ are independent of all other stochastic processes. This implies that

$$(1.17) \quad \rho_{12} = c\rho_{22}, \quad \rho_{21} = c\rho_{11}, \quad r = c\rho_{11}\rho_{22}.$$

As in the “single stochastic vol” case, there is only a single parameter, the FX rate correlation c , in addition to the parameters $\alpha_1, \rho_{11}, \nu_1$ and $\alpha_2, \rho_{22}, \nu_2$ determined by the component FX rate volatility smiles.

Substituting 1.17 into 1.13a-1.15e yields the effective SABR parameters for this case. This doesn’t lead to any significant simplifications, so we haven’t written these formulas explicitly.

2. Other crosses. There are liquid option markets for some cross rates $\tilde{X}_{ij}$. For some of these, the liquid options for the component FX rates are for the rates $\tilde{X}_{ik}$, $\tilde{X}_{kj}$, so that

$$(2.1) \quad \tilde{X}_{ij} = \tilde{X}_{ik}\tilde{X}_{kj},$$

instead of $\tilde{X}_{ij} = \tilde{X}_{ik}/\tilde{X}_{jk}$. If the log normal SABR model is used to model these FX rates, so

$$(2.2a) \quad d\tilde{X}_{ik} = \varepsilon\tilde{A}_1\tilde{X}_{ik}dW_1,$$

$$(2.2b) \quad d\tilde{A}_1 = \varepsilon\nu_1\tilde{A}_1dZ_1,$$

$$(2.2c) \quad dW_1dZ_1 = \rho_{11}dT,$$

in the forward measure for currency k , and

$$(2.3a) \quad d\tilde{X}_{kj} = \varepsilon\tilde{A}_2\tilde{X}_{kj}dW_2,$$

$$(2.3b) \quad d\tilde{A}_2 = \varepsilon\nu_2\tilde{A}_2dZ_2,$$

$$(2.3c) \quad dW_2dZ_2 = \rho_{22}dT,$$

in the forward measure for currency j , then Itô's formula yields

$$(2.4) \quad d\tilde{X}_{ij} = d\left(\tilde{X}_{ik}\tilde{X}_{kj}\right) = \tilde{X}_{kj}d\tilde{X}_{ik} + \tilde{X}_{ik}d\tilde{X}_{kj} + \text{drift terms}.$$

Using the forward measure for currency j eliminates the drift terms, so

$$(2.5a) \quad d\tilde{X}_{ij} = \varepsilon\tilde{X}_{ij}\left(\tilde{A}_1dW_1 + \tilde{A}_2dW_2\right),$$

$$(2.5b) \quad d\tilde{A}_1 = \varepsilon\nu_1\tilde{A}_1dZ_1,$$

$$(2.5c) \quad d\tilde{A}_2 = \varepsilon\nu_2\tilde{A}_2dZ_2,$$

with $dW_1dZ_1 = \rho_{11}dT$ and $dW_2dZ_2 = \rho_{22}dT$. This is identical to eqs.1.5a-1.5c in the previous section, except that $-\tilde{A}_2dW_2$ is now $+\tilde{A}_2dW_2$.

Similarly, the reverse cross is $\tilde{X}_{ji} = \left(1/\tilde{X}_{ij}\right)$, so Itô's lemma yields

$$(2.6) \quad d\tilde{X}_{ji} = -\frac{d\tilde{X}_{ij}}{\tilde{X}_{ij}^2} + \text{drift terms}.$$

In the appropriate measure for this FX rate (the forward measure for currency i) the drift terms are identically zero, so

$$(2.7a) \quad d\tilde{X}_{ji} = \varepsilon\tilde{X}_{ji}\left(-\tilde{A}_1dW_1 - \tilde{A}_2dW_2\right),$$

$$(2.7b) \quad d\tilde{A}_1 = \varepsilon\nu_1\tilde{A}_1dZ_1,$$

$$(2.7c) \quad d\tilde{A}_2 = \varepsilon\nu_2\tilde{A}_2dZ_2.$$

This too is identical to eqs. 1.5a-1.5c, but $\tilde{A}_1dW_1$ is now $-\tilde{A}_1dW_1$.

In Appendices A and B we handle all these cases by analyzing

$$(2.8a) \quad d\tilde{X}_{ij} = \varepsilon\tilde{X}_{ij}\left(\lambda_1\tilde{A}_1dW_1 + \lambda_2\tilde{A}_2dW_2\right),$$

$$(2.8b) \quad d\tilde{A}_1 = \varepsilon\nu_1\tilde{A}_1dZ_1,$$

$$(2.8c) \quad d\tilde{A}_2 = \varepsilon\nu_2\tilde{A}_2dZ_2,$$

where λ_1, λ_2 are constant. There it is shown that the probability distribution for the cross FX rate $\tilde{X}_{ij}$, and thus the prices of European options on $\tilde{X}_{ij}$, can be obtained from the SABR model 1.9a-1.9c to within $O(\varepsilon^2)$. For arbitrary correlations

$$(2.9a) \quad dW_1 dW_2 = c dT, \quad dW_1 dZ_1 = \rho_{11} dT, \quad dW_2 dZ_2 = \rho_{22} dT,$$

$$(2.9b) \quad dW_1 dZ_2 = \rho_{12} dT, \quad dW_2 dZ_1 = \rho_{21} dT, \quad dZ_1 dZ_2 = r dT,$$

the effective SABR parameters are given by⁴ B.5a-B.5c with B.3a-B.3d and B.6a-B.6e. For the special case of a single stochastic vol $dZ_1 = dZ_2 = dZ$, so that

$$(2.10) \quad \rho_{12} = \rho_{11}, \quad \rho_{21} = \rho_{22}, \quad r = 1,$$

these coefficients simplify to B.5a-B.5c with B.7a-B.8c.

3. Conclusions. Most currencies i have well-developed option markets for currency i /USD FX rates. By using SABR models to model these rates, the preceding results allow us to derive SABR models for all the cross FX rates $\tilde{X}_{ij}$ from the SABR models for the currency i /USD and currency j /USD FX rates. That is, it determines the volatility smile of the cross FX rates from the smiles of their component FX rates. If there is an active option market for the cross FX rate, these results determine if the cross FX volatility smile is consistent with the volatilities of the component FX rates. If not, we can synthetically re-create the cross FX options by dynamically hedging with options on the component FX rates, monetizing the difference between the synthetic and real option prices. If there isn't an active market, we can inject liquidity into the cross FX options market, using these results to hedge our risks in the component FX rate option markets.

In section 1 we used a single stochastic process to model the currency i /base FX rate,

$$(3.1) \quad d\tilde{X}_{i0} = \varepsilon \tilde{X}_{i0} \tilde{A}_i dW_i.$$

Modeled this way, the factor dW_i must reflect events in both economies. If the economy of the base currency weakens or strengthens, most of the base FX rates dW_i would increase or decrease together. I.e., the common dependence on the base currency leads to excessive dependencies between the dW_i 's. We could potentially eliminate much of this cross correlation if we had a factor dW_0 to model the events in the base economy. This suggests a new model,

$$(3.2) \quad d\tilde{X}_{i0} = \varepsilon \tilde{X}_{i0} \left\{ \tilde{A}_i dW_i - \tilde{A}_0 dW_0 \right\} \quad \text{for all } i,$$

for FX rates into the base currency. Thus, *all* FX rates would be modeled as

$$(3.3a) \quad d\tilde{X}_{ij} = \varepsilon \tilde{X}_{ij} \left\{ \tilde{A}_i dW_i - \tilde{A}_j dW_j \right\},$$

$$(3.3b) \quad d\tilde{A}_i = \varepsilon \nu_i \tilde{A}_i dZ_i,$$

$$(3.3c) \quad d\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j dZ_j,$$

for $i = 0, \dots, n$, with

$$(3.3d) \quad dW_i dW_j = c_{ij} dT, \quad dZ_i dZ_j = r_{ij} dT,$$

$$(3.3e) \quad dW_i dZ_j = \rho_{ij} dT.$$

Note that we are omitting the drift term throughout because using the appropriate numeraire (the zero coupon bond in currency j for the FX rate $\tilde{X}_{ij}$) means that the drift rates are identically zero.

⁴Note that in these formulas, $a_1 = \lambda_1 \alpha_1$ and $a_2 = \lambda_2 \alpha_2$.

It is easily seen that this model is self-consistent across all currencies. The reverse FX rate is

$$(3.4) \quad \tilde{X}_{ji} = \frac{\text{ccy } j}{\text{ccy } i} = \text{forward price of 1 unit of ccy } j \text{ in terms of ccy } i,$$

and using Itô's formula yields

$$(3.5) \quad d\tilde{X}_{ji} = d\left(1/\tilde{X}_{ij}\right) = \varepsilon\tilde{X}_{ji} \left\{ \tilde{A}_j dW_j - \tilde{A}_i dW_i \right\}.$$

This is identical to 3.3a with $i \leftrightarrow j$. Similarly, for any three currencies,

$$(3.6) \quad \frac{\tilde{X}_{ik}}{\tilde{X}_{jk}} = \tilde{X}_{ij} \quad \text{for any ccy's } i, j, k.$$

Using Itô's formula and 3.3a yields

$$(3.7) \quad \begin{aligned} d\tilde{X}_{ij} &= d\left(\frac{\tilde{X}_{ik}}{\tilde{X}_{jk}}\right) = \varepsilon\frac{\tilde{X}_{ik}}{\tilde{X}_{jk}} \left\{ \tilde{A}_i dW_i - \tilde{A}_k dW_k \right\} - \varepsilon\frac{\tilde{X}_{ik}}{\tilde{X}_{jk}} \left\{ \tilde{A}_j dW_j - \tilde{A}_k dW_k \right\} \\ &= \varepsilon\tilde{X}_{ij} \left\{ \tilde{A}_i dW_i - \tilde{A}_j dW_j \right\}, \end{aligned}$$

so we again recover 3.3a. In the same way,

$$(3.8) \quad \tilde{X}_{ik}\tilde{X}_{kj} = \tilde{X}_{ij} \quad \text{for any ccy's } i, j, k,$$

and again

$$(3.9) \quad \begin{aligned} d\tilde{X}_{ij} &= d\left(\tilde{X}_{ik}\tilde{X}_{kj}\right) = \varepsilon\tilde{X}_{ik}\tilde{X}_{kj} \left\{ \tilde{A}_i dW_i - \tilde{A}_k dW_k \right\} + \varepsilon\tilde{X}_{ik}\tilde{X}_{kj} \left\{ \tilde{A}_k dW_k - \tilde{A}_j dW_j \right\} \\ &= \varepsilon\tilde{X}_{ij} \left\{ \tilde{A}_i dW_i - \tilde{A}_j dW_j \right\}. \end{aligned}$$

Although the extra factor, dW_0 , appears to be a complication, it may simplify the model in practice. First, we can always get back to the preliminary model by setting $\tilde{A}_0$ to zero. (Since $\tilde{A}_0$ is log normal, it suffices to set $\tilde{A}_0(0) = 0$). Second, analyzing the new model to obtain the smile of a primary FX rate $\tilde{X}_{i0}$ is identical to the problem of obtaining the smiles for cross rates $\tilde{X}_{ij}$. So we already have explicit implied volatility formulas for these models. Third, the stochastic processes dW_i , dZ_i now reflect events in economy i and dW_j , dZ_j encapsulate events in economy j . Any remaining correlations between dW_i and dW_j now represents correlations between the two economies of these currencies. We believe that such correlations should be more stable, more intuitive, and significantly smaller than correlations that need to reflect strengthening and weakening of both the currency and the base currency. Finally, in this model all the FX rates are governed by the same model.

Recently, “mixture” models have been proposed for many asset types, where the underlying is driven by several stochastic processes. For example, they can arise when econometric and data analytic approaches are used to create fundamental economic models. So consider FX models of the form

$$(3.10a) \quad d\tilde{X}_{ij} = \varepsilon\tilde{X}_{ij} \left\{ \sum_m \tilde{A}_{ij}^{(m)} dW_m \right\},$$

$$(3.10b) \quad d\tilde{A}_{ij}^{(m)} = \varepsilon\nu_{ij}^{(m)} \tilde{A}_{ij}^{(m)} dZ_{ij}^{(m)},$$

with

$$(3.10c) \quad dW_m dW_n = c_{mn} dT, \quad dW_m dZ_{ij}^{(n)} = \rho_{ij}^{mn} dT, \quad dZ_{ij}^{(m)} dZ_{k\ell}^{(n)} = r_{ij,k\ell}^{mn}.$$

Since $\tilde{X}_{ij} = \tilde{X}_{ik}/\tilde{X}_{jk}$ for any currency k , consistency between the FX rates requires that

$$(3.11) \quad d\tilde{X}_{ij} = d\left(\frac{\tilde{X}_{ik}}{\tilde{X}_{jk}}\right) = \varepsilon \tilde{X}_{ij} \sum_m \left(\tilde{A}_{ik}^{(m)} - \tilde{A}_{jk}^{(m)}\right) dW_m,$$

for any k , so clearly we must require

$$(3.12) \quad \tilde{A}_{ij}^{(m)} = \tilde{A}_{ik}^{(m)} - \tilde{A}_{jk}^{(m)} \quad \text{for any } k.$$

Without loss of generality, let us choose ccy 0 to be the base currency. This base currency is usually chosen to be USD, but it could equally well be any other currency. To simplify notation, define

$$(3.13a) \quad \tilde{X}_i = \tilde{X}_{i0},$$

$$(3.13b) \quad \tilde{A}_i^{(m)} = \tilde{A}_{i0}^{(m)}.$$

Then the FX rates into the base currency are

$$(3.14a) \quad d\tilde{X}_i = \varepsilon \tilde{X}_i \sum_m \tilde{A}_i^{(m)} dW_m,$$

and the cross rates are now

$$(3.14b) \quad d\tilde{X}_{ij} = \varepsilon \tilde{X}_{ij} \sum_m \left(\tilde{A}_i^{(m)} - \tilde{A}_j^{(m)}\right) dW_m,$$

where

$$(3.14c) \quad d\tilde{A}_i^{(m)} = \varepsilon \nu_i^{(m)} \tilde{A}_i^{(m)} dZ_i^{(m)},$$

and

$$(3.14d) \quad dW_m dW_n = c_{mn} dT, \quad dZ_i^{(m)} dZ_j^{(n)} = r_{ij}^{mn} dT,$$

$$(3.14e) \quad dW_m dZ_i^{(n)} = \rho_i^{mn} dT.$$

To handle these multi-factor models, in Appendix A we analyze models of the form

$$(3.15a) \quad d\tilde{X} = \varepsilon \tilde{X} \sum_{j=1}^n \lambda_j \tilde{A}_j dW_j,$$

$$(3.15b) \quad d\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j dZ_j,$$

where the λ_j are constants. We find that the probability distribution and option prices are determined by an effective SABR model through $O(\varepsilon^2)$. We then specialize the results to $n = 2$ in Appendix B; recovering the cross FX models by setting $\lambda_1 = 1$ and $\lambda_2 = -1$.

Appendix A. Derivation of the effective forward equation.

Let $\tilde{X}$ be the forward FX rate for any specific settlement date T_{set} . We use singular perturbation methods to analyze the general stochastic vol model,

$$(A.1a) \quad d\tilde{X} = \varepsilon \tilde{X} \sum_{j=1}^n \lambda_j \tilde{A}_j dW_j,$$

$$(A.1b) \quad d\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j dZ_j,$$

in the asymptotic limit $\varepsilon \ll 1$, where we allow a full correlation structure:

$$(A.1c) \quad dW_i dW_j = c_{ij} dT, \quad dW_i dZ_j = \rho_{ij} dT, \quad dZ_i dZ_j = r_{ij} dT.$$

A.1. Conservation law. If the volatilities weren't stochastic, so the $\tilde{A}_j$'s were constant, then we could combine the Brownian motions into a single Brownian motion

$$(A.2a) \quad d\tilde{X} = \varepsilon \Delta(\mathbf{A}) \tilde{X} dW,$$

where

$$(A.2b) \quad \Delta^2(\mathbf{A}) = \sum_{i,j} c_{ij} \lambda_i \lambda_j A_i A_j.$$

So we expect $\Delta(\mathbf{A})$ to be the leading order volatility for $\tilde{X}$ when $\varepsilon \ll 1$.

Define the probability density

$$(A.3) \quad p(t, x, \boldsymbol{\alpha}; T, X, \mathbf{A}) \equiv \mathbb{E} \left\{ \delta \left(\tilde{X}(T) - X \right) \delta^n \left(\tilde{\mathbf{A}}(T) - \mathbf{A} \right) \middle| \tilde{X}(t) = x, \tilde{\mathbf{A}}(t) = \boldsymbol{\alpha} \right\},$$

and the moments

$$(A.4) \quad \begin{aligned} Q^{(k)}(t, x, \boldsymbol{\alpha}; T, X) &= \mathbb{E} \left\{ \Delta^k(\tilde{\mathbf{A}}(T)) \delta \left(\tilde{X}(T) - X \right) \middle| \tilde{X}(t) = x, \tilde{\mathbf{A}}(t) = \boldsymbol{\alpha} \right\} \\ &= \int \cdots \int \Delta^k(\mathbf{A}) p(t, x, \boldsymbol{\alpha}; T, X, \mathbf{A}) dA_1 \cdots dA_n. \end{aligned}$$

The forward Kolmogorov (Fökker-Planck) equation for the density is

$$(A.5a) \quad p_T = \frac{1}{2} \varepsilon^2 \left\{ X^2 \Delta^2(\mathbf{A}) p \right\}_{XX} + \varepsilon^2 \sum_{i,j} \lambda_i \rho_{ij} \nu_j (A_i A_j X p)_{XA_j} + \frac{1}{2} \varepsilon^2 \sum_{i,j} r_{ij} \nu_i \nu_j (A_i A_j p)_{A_i A_j}$$

for $T > t$, with the initial condition

$$(A.5b) \quad p \rightarrow \delta(X - x) \delta^n(\mathbf{A} - \boldsymbol{\alpha}) \quad \text{as } T \rightarrow t.$$

The second and third terms are perfect derivatives with respect to A_j , so integrating over all A_j yields zero:

$$(A.6) \quad \int (A_i A_j X p)_{XA_j} dA_j = 0, \quad \int (A_i A_j p)_{A_i A_j} dA_j = 0.$$

Consequently, if we integrate the forward Kolmogorov equation over all the A 's, we obtain

$$(A.7a) \quad Q_T^{(0)} = \frac{1}{2} \varepsilon^2 \left\{ X^2 Q^{(2)} \right\}_{XX} \quad \text{for } T > t,$$

$$(A.7b) \quad Q^{(0)}(T, X) \rightarrow \delta(X - x) \quad \text{as } T \rightarrow t.$$

The marginal density

$$(A.8) \quad Q^{(0)}(t, x, \boldsymbol{\alpha}; T, X) = \int \cdots \int p(t, x, \boldsymbol{\alpha}; T, X, \mathbf{A}) dA_1 \cdots dA_n$$

is the probability that the FX rate is X , regardless of the values of the volatilities. Equation A.7a is a conservation equation for this marginal density. In the next section we use asymptotic methods to analyze the *backwards* Kolmogorov equation for the moments $Q^{(k)}$. This analysis establishes a “constitutive” relationship that determines $Q^{(2)}$ in terms of $Q^{(0)}$,

$$(A.9a) \quad Q^{(2)} = \Delta^2(\boldsymbol{\alpha}) e^{\varepsilon^2(\Gamma + \beta\Delta)(T-t)} (1 + 2\varepsilon\beta z + \varepsilon^2\gamma z^2) Q^{(0)},$$

where

$$(A.9b) \quad z = \frac{1}{\varepsilon\Delta(\boldsymbol{\alpha})} \log X/x.$$

To write the coefficients β , γ , Γ succinctly, we set

$$(A.10) \quad a_i = \lambda_i \alpha_i \quad \text{for all } i.$$

Then these coefficients will turn out to be

$$(A.11a) \quad \Delta^2(\boldsymbol{\alpha}) = \sum_{i,j} c_{ij} a_i a_j,$$

$$(A.11b) \quad \beta(\boldsymbol{\alpha}) = \frac{1}{\Delta^3} \sum_{i,j,k} \rho_{ij} \nu_j c_{jk} a_i a_j a_k,$$

$$(A.11c) \quad \gamma(\boldsymbol{\alpha}) = \frac{1}{\Delta^4} \sum_{i,j,k,m} r_{ij} \nu_i \nu_j c_{ik} c_{jm} a_i a_j a_k a_m + \kappa(\boldsymbol{\alpha}),$$

$$(A.11d) \quad \Gamma(\boldsymbol{\alpha}) = \frac{1}{\Delta^4} \sum_{i,j,k,m} r_{ij} \nu_i \nu_j (c_{ij} c_{km} - c_{ik} c_{jm}) a_i a_j a_k a_m - \kappa(\boldsymbol{\alpha}),$$

$$(A.11e) \quad \kappa(\boldsymbol{\alpha}) = \frac{1}{\Delta^4} \sum_{i,j,k,m} \rho_{ij} \nu_j (\rho_{ki} \nu_i + \rho_{kj} \nu_j + \rho_{km} \nu_m) c_{jm} a_i a_j a_k a_m - 3\beta^2(\boldsymbol{\alpha}).$$

See A.34a-A.34f. Although not exact, this relationship is accurate through $O(\varepsilon^2)$, the same accuracy as the original SABR formulas. Substituting this relationship into the conservation law then yields the *effective forward equation*. Without loss of generality, we set the initial time $t = 0$. The effective forward equation for the marginal density $Q^{(0)}(T, X) \equiv Q^{(0)}(0, x, \boldsymbol{\alpha}; T, X)$ is then

$$(A.12a) \quad Q_T^{(0)} = \frac{1}{2} \varepsilon^2 \Delta^2 e^{\varepsilon^2(\Gamma + \beta\Delta)T} \left[(1 + 2\varepsilon\beta z + \varepsilon^2\gamma z^2) X^2 Q^{(0)} \right]_{XX} \quad \text{for } T > 0,$$

$$(A.12b) \quad Q^{(0)}(T, X) \rightarrow \delta(X - x) \quad \text{as } T \rightarrow 0^+.$$

Even though the constitutive law is not exact, *option prices obtained from the conservation law are exactly arbitrage free*, since the maximum principle[8] ensures that the probability density $Q^{(0)}$ is strictly positive for all $T > 0$. So instead of viewing the effective forward equation as an asymptotic solution, it can be regarded as an arbitrage free model in its own right, a model that is asymptotically close to A.1a-A.1c.

After deriving the effective forward equation, we re-express these results as an effective SABR model. Specifically, the analysis in [2] is used to show that for any expiry date T_{ex} , the probability density

$Q^{(0)}(T_{ex}, X)$ obtained from the effective forward equation A.12a, A.12b and the probability density obtained from the log normal SABR model

$$\begin{aligned} \text{(A.13a)} \quad & d\tilde{X} = \varepsilon \tilde{A} \tilde{X} dW, \\ \text{(A.13b)} \quad & d\tilde{A} = \varepsilon \nu_{eff} \tilde{A} dZ, \\ \text{(A.13c)} \quad & dZ dW = \rho_{eff} dT, \end{aligned}$$

are identical to within $O(\varepsilon^2)$, provided that the SABR parameters $\alpha_{eff} = \tilde{A}(0)$, ρ_{eff} , and ν_{eff} are chosen to be

$$\begin{aligned} \text{(A.14a)} \quad & \alpha^{eff} = \Delta(\alpha) e^{\frac{1}{4}\varepsilon^2 \Gamma(\alpha) T_{ex}}, \\ \text{(A.14b)} \quad & \rho^{eff} = \beta(\alpha) / \sqrt{\gamma(\alpha)}, \\ \text{(A.14c)} \quad & \nu^{eff} = \sqrt{\gamma(\alpha)}. \end{aligned}$$

Consequently, this SABR model can be used to price all European options with exercise date T_{ex} . These prices are not exact, but they are accurate to within $O(\varepsilon^2)$, the same accuracy as the original SABR formulas⁵. Thus, standard European options[5] or quadratic options[6] can be valued either by using the SABR model's explicit implied volatility formulas contained in [5] and [6], or by numerically solving the effective forward equation[2].

A.2. The backwards equation. Let

$$\text{(A.15)} \quad \tau = T - t.$$

The backwards Kolmogorov equation for the moments $Q^{(k)}$ is

$$\text{(A.16a)} \quad Q_\tau^{(k)} = \frac{1}{2}\varepsilon^2 \Delta^2 x^2 Q_{xx}^{(k)} + \varepsilon^2 \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j x Q_{x\alpha_j}^{(k)} + \frac{1}{2}\varepsilon^2 \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j Q_{\alpha_i \alpha_j}^{(k)}$$

for $\tau > 0$, with

$$\text{(A.16b)} \quad Q^{(k)} \rightarrow \Delta^k(\alpha) \delta(X - x) \quad \text{as } \tau \rightarrow 0.$$

We proceed by repeatedly changing variables until the relationship between $Q^{(2)}$ and $Q^{(0)}$ is obvious. Let

$$\text{(A.17)} \quad z = \frac{\ln X/x}{\varepsilon \Delta(\alpha)}.$$

⁵Note that a different α^{eff} needs to be used for each different exercise date T_{ex} . Alternatively, the SABR model with

$$d\tilde{X} = \varepsilon \tilde{A} \tilde{X} e^{\frac{1}{2}\varepsilon^2 \Gamma(\alpha) T} dW,$$

and $\alpha^{eff} \equiv \tilde{A}(0) = \Delta(\alpha)$ yields the probability density $Q^{(0)}(T, X)$ to within $O(\varepsilon^2)$ for all T , but at the cost of having a time-dependent coefficient in our SABR model.

Changing variables from x, α to z, α requires

$$(A.18a) \quad \frac{\partial}{\partial x} \rightarrow -\frac{1}{\varepsilon \Delta x} \frac{\partial}{\partial z}, \quad \frac{\partial^2}{\partial x^2} \rightarrow \frac{1}{\varepsilon^2 \Delta^2 x^2} \left\{ \frac{\partial^2}{\partial z^2} + \varepsilon \Delta \frac{\partial}{\partial z} \right\},$$

$$(A.18b) \quad \frac{\partial}{\partial \alpha_i} \rightarrow \frac{\partial}{\partial \alpha_i} - \frac{\Delta_{\alpha_i}}{\Delta} z \frac{\partial}{\partial z},$$

$$(A.18c) \quad \frac{\partial^2}{\partial x \partial \alpha_i} \rightarrow \frac{1}{\varepsilon \Delta x} \left\{ -\frac{\partial^2}{\partial z \partial \alpha_i} + \frac{\Delta_{\alpha_i}}{\Delta} z \frac{\partial^2}{\partial z^2} + \frac{\Delta_{\alpha_i}}{\Delta} \frac{\partial}{\partial z} \right\},$$

$$(A.18d) \quad \begin{aligned} \frac{\partial^2}{\partial \alpha_i \alpha_j} \rightarrow & \frac{\partial^2}{\partial \alpha_i \alpha_j} - \frac{\Delta_{\alpha_j}}{\Delta} z \frac{\partial^2}{\partial z \partial \alpha_i} - \frac{\Delta_{\alpha_i}}{\Delta} z \frac{\partial^2}{\partial z \partial \alpha_j} \\ & + \frac{\Delta_{\alpha_i} \Delta_{\alpha_j}}{\Delta^2} \left\{ z^2 \frac{\partial^2}{\partial z^2} + 2z \frac{\partial}{\partial z} \right\} - \frac{\Delta_{\alpha_i \alpha_j}}{\Delta} z \frac{\partial}{\partial z}. \end{aligned}$$

Finally,

$$(A.18e) \quad \delta(X - x) \rightarrow \frac{\delta(z)}{\varepsilon \Delta(\alpha) X}.$$

In the new variables, the backwards equation is

$$(A.19a) \quad \begin{aligned} Q_\tau^{(k)} = & \frac{1}{2} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) Q_{zz}^{(k)} + \frac{1}{2} \varepsilon \Delta Q_z^{(k)} + \varepsilon \beta Q_z^{(k)} + \varepsilon^2 (g - \frac{1}{2} \phi) z Q_z^{(k)} \\ & - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j Q_{z\alpha_j}^{(k)} + \frac{1}{2} \varepsilon^2 \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j \left\{ Q_{\alpha_i \alpha_j}^{(k)} - 2 \frac{\Delta_{\alpha_i}}{\Delta} z Q_{z\alpha_j}^{(k)} \right\} \end{aligned}$$

for $\tau > 0$, with the initial condition

$$(A.19b) \quad Q^{(k)} \rightarrow \frac{\Delta^{k-1}}{\varepsilon X} \delta(z) \quad \text{as } \tau \rightarrow 0.$$

Here,

$$(A.20a) \quad \beta(\alpha) = \frac{1}{\Delta^2} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j \Delta_{\alpha_j},$$

$$(A.20b) \quad g(\alpha) = \frac{1}{\Delta^2} \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j \Delta_{\alpha_i} \Delta_{\alpha_j},$$

$$(A.20c) \quad \phi(\alpha) = \frac{1}{\Delta} \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j \Delta_{\alpha_i \alpha_j}.$$

Next, define $S^{(k)}$ via

$$(A.21) \quad Q^{(k)} = \frac{1}{\varepsilon} \frac{\Delta^{k-1}(\alpha)}{X} S^{(k)}.$$

Then

$$(A.22a) \quad Q_{z\alpha_i}^{(k)} \rightarrow S_{z\alpha_i}^{(k)} + (k-1) \frac{\Delta_{\alpha_i}}{\Delta} S_z^{(k)},$$

$$(A.22b) \quad Q_{\alpha_i \alpha_j}^{(k)} \rightarrow S_{\alpha_i \alpha_j}^{(k)} + \frac{k-1}{\Delta} \left(\Delta_{\alpha_i} S_{\alpha_j}^{(k)} + \Delta_{\alpha_j} S_{\alpha_i}^{(k)} + \Delta_{\alpha_i \alpha_j} S^{(k)} \right) + \frac{(k-1)(k-2)}{\Delta^2} \Delta_{\alpha_i} \Delta_{\alpha_j} S^{(k)}.$$

The backwards equation now becomes

$$\begin{aligned}
(\text{A.23a}) \quad S_\tau^{(k)} &= \frac{1}{2} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) S_{zz}^{(k)} + \frac{1}{2}\varepsilon\Delta S_z^{(k)} - \varepsilon (k-2) \beta S_z^{(k)} - \varepsilon^2 (k-2) g z S_z^{(k)} \\
&\quad + \frac{1}{2}\varepsilon^2 (k-1) (k-2) g S^{(k)} - \frac{1}{2}\varepsilon^2 \phi \left\{ z S_z^{(k)} - (k-1) S^{(k)} \right\} - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j S_{z\alpha_j}^{(k)} \\
&\quad + \frac{1}{2}\varepsilon^2 \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j \left\{ S_{\alpha_i \alpha_j}^{(k)} + 2(k-1) \frac{\Delta_{\alpha_i}}{\Delta} S_{\alpha_j}^{(k)} - 2 \frac{\Delta_{\alpha_i}}{\Delta} z S_{z\alpha_j}^{(k)} \right\}
\end{aligned}$$

through $O(\varepsilon^2)$, where

$$(\text{A.23b}) \quad S^{(k)} \rightarrow \delta(z) \quad \text{as } \tau \rightarrow 0.$$

Examining this equation shows that the equation and initial condition for $S^{(k)}$ is independent of α to leading order. Therefore, $S^{(k)}$ is independent of α to leading order. This means that $\varepsilon^2 S_{\alpha_i \alpha_j}^{(k)}$, $\varepsilon^2 S_{\alpha_j}^{(k)}$, and $\varepsilon^2 S_{z\alpha_j}^{(k)}$ are all no larger than $O(\varepsilon^3)$. As we are only working through $O(\varepsilon^2)$, we neglect these terms, obtaining

$$\begin{aligned}
(\text{A.24a}) \quad S_\tau^{(k)} &= \frac{1}{2} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) S_{zz}^{(k)} + \frac{1}{2}\varepsilon\Delta S_z^{(k)} - \varepsilon (k-2) \beta S_z^{(k)} - \varepsilon^2 (k-2) g z S_z^{(k)} \\
&\quad + \frac{1}{2}\varepsilon^2 (k-1) (k-2) g S^{(k)} - \frac{1}{2}\varepsilon^2 \phi \left\{ z S_z^{(k)} - (k-1) S^{(k)} \right\} - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j S_{z\alpha_j}^{(k)},
\end{aligned}$$

$$(\text{A.24b}) \quad S^{(k)} \rightarrow \delta(z) \quad \text{as } \tau \rightarrow 0.$$

For the special cases of $k=0$ and $k=2$, this yields

$$\begin{aligned}
(\text{A.25a}) \quad S_\tau^{(0)} &= \frac{1}{2} \left[(1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) S_{zz}^{(0)} + \frac{1}{2}\varepsilon\Delta S_z^{(0)} - \frac{1}{2}\varepsilon^2 \phi \left\{ z S_z^{(0)} + S^{(0)} \right\} \right. \\
&\quad \left. - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j S_{z\alpha_j}^{(0)} \right],
\end{aligned}$$

$$\begin{aligned}
(\text{A.25b}) \quad S_\tau^{(2)} &= \frac{1}{2} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) S_{zz}^{(2)} + \frac{1}{2}\varepsilon\Delta S_z^{(2)} - \frac{1}{2}\varepsilon^2 \phi \left\{ z S_z^{(2)} - S^{(2)} \right\} \\
&\quad - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j S_{z\alpha_j}^{(2)},
\end{aligned}$$

$$(\text{A.25c}) \quad S^{(0)} \rightarrow \delta(z), \quad S^{(2)} \rightarrow \delta(z) \quad \text{as } \tau \rightarrow 0.$$

The equations for $S^{(2)}$ and $S^{(0)}$ are very similar; we now transform $S^{(0)}$ until they are identical, at least through $O(\varepsilon^2)$. Define

$$(\text{A.26}) \quad U = (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) S^{(0)}.$$

Equation A.25a becomes

$$\begin{aligned}
(\text{A.27a}) \quad U_\tau &= \frac{1}{2} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) U_{zz} + \frac{1}{2}\varepsilon\Delta U_z - \frac{1}{2}\varepsilon^2 \phi \{ z U_z - U \} \\
&\quad - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j U_{z\alpha_j} + 2\varepsilon^2 \kappa (z U)_z - \varepsilon^2 (\Delta\beta + \phi) U,
\end{aligned}$$

$$(\text{A.27b}) \quad U \rightarrow \delta(z) \quad \text{as } \tau \rightarrow 0,$$

through $O(\varepsilon^2)$, where we have neglected the terms with $\varepsilon^2 U_{\alpha_j}$ as they are actually $O(\varepsilon^3)$ or smaller. Here

$$(\text{A.28}) \quad \kappa = \frac{1}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j \beta_{\alpha_j}.$$

Finally, we set

$$(A.29) \quad U = H e^{-\varepsilon^2 \kappa z^2 - \varepsilon^2 (\beta \Delta + \phi - \kappa) \tau}.$$

In terms of H , we obtain

$$(A.30a) \quad H_\tau = \frac{1}{2} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) H_{zz} - \frac{1}{2} \varepsilon^2 \phi \{z H_z - H\} - \frac{\varepsilon}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j H_{z\alpha_j},$$

$$(A.30b) \quad H \rightarrow \delta(z) \quad \text{as } \tau \rightarrow 0,$$

through $O(\varepsilon^2)$. These equations are identical to equations A.25b, A.25c for $S^{(2)}$. Therefore, H is identical to $S^{(2)}$, at least through $O(\varepsilon^2)$. Tracking back through the transformations, we see that this means that

$$(A.31a) \quad \begin{aligned} Q^{(2)} &= \Delta^2(\boldsymbol{\alpha}) e^{\varepsilon^2 \kappa z^2 + \varepsilon^2 (\beta \Delta + \Gamma) \tau} (1 + 2\varepsilon\beta z + \varepsilon^2 g z^2) Q^{(0)} \\ &\equiv \Delta^2(\boldsymbol{\alpha}) e^{\varepsilon^2 (\Gamma + \beta \Delta) \tau} (1 + 2\varepsilon\beta z + \varepsilon^2 \gamma z^2) Q^{(0)}, \end{aligned}$$

through $O(\varepsilon^2)$, where

$$(A.31b) \quad \gamma = g + \kappa, \quad \Gamma = \phi - \kappa.$$

A.3. The effective forward equation. Substituting the constitutive law A.31a, A.31b into our conservation equation A.7a yields the *effective forward equation* for the marginal density. Without loss of generality, we set the initial time $t = 0$. Then the effective forward equation for $Q^{(0)}(T, X) \equiv Q^{(0)}(0, x, \boldsymbol{\alpha}; T, X)$ is

$$(A.32a) \quad Q_T^{(0)} = \frac{1}{2} \varepsilon^2 \Delta^2(\boldsymbol{\alpha}) e^{\varepsilon^2 (\beta \Delta + \Gamma) T} \left[(1 + 2\varepsilon\beta z + \varepsilon^2 \gamma z^2) X^2 Q^{(0)} \right]_{XX} \quad \text{for } T > 0,$$

$$(A.32b) \quad Q^{(0)}(T, X) \rightarrow \delta(X - x) \quad \text{as } T \rightarrow 0^+.$$

To express the coefficients more succinctly, we set

$$(A.33) \quad a_i = \lambda_i \alpha_i \quad \text{for all } i.$$

Then,

$$(A.34a) \quad z = \frac{\ln X/x}{\varepsilon \Delta(\boldsymbol{\alpha})}, \quad \gamma(\boldsymbol{\alpha}) = g(\boldsymbol{\alpha}) + \kappa(\boldsymbol{\alpha}), \quad \Gamma(\boldsymbol{\alpha}) = \phi(\boldsymbol{\alpha}) - \kappa(\boldsymbol{\alpha}),$$

with

$$(A.34b) \quad \Delta^2(\alpha) = \sum_{i,j} c_{ij} a_i a_j,$$

$$(A.34c) \quad \beta(\alpha) = \frac{1}{\Delta^2} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j \Delta_{\alpha_j} = \frac{1}{\Delta^3} \sum_{i,j,k} \rho_{ij} \nu_j c_{jk} a_i a_j a_k,$$

$$(A.34d) \quad g(\alpha) = \frac{1}{\Delta^2} \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j \Delta_{\alpha_i} \Delta_{\alpha_j} \\ = \frac{1}{\Delta^4} \sum_{i,j,k,m} r_{ij} \nu_i \nu_j c_{ik} c_{jm} a_i a_j a_k a_m,$$

$$(A.34e) \quad \phi(\alpha) = \frac{1}{\Delta} \sum_{i,j} r_{ij} \nu_i \nu_j \alpha_i \alpha_j \Delta_{\alpha_i \alpha_j} \\ = \frac{1}{\Delta^4} \sum_{i,j,k,m} r_{ij} \nu_i \nu_j (c_{ij} c_{km} - c_{ik} c_{jm}) a_i a_j a_k a_m,$$

$$(A.34f) \quad \kappa(\alpha) = \frac{1}{\Delta} \sum_{i,j} \lambda_i \rho_{ij} \nu_j \alpha_i \alpha_j \beta_{\alpha_j} \\ = \frac{1}{\Delta^4} \sum_{i,j,k,m} \rho_{ij} \nu_j (\rho_{ki} \nu_i + \rho_{kj} \nu_j + \rho_{km} \nu_m) c_{jm} a_i a_j a_k a_m - 3\beta^2(\alpha).$$

A.4. The effective SABR model. The effective forward equation A.32a, A.32b and coefficients A.34a-A.34f allow European options to be priced by solving a one spatial dimension PDE. We now re-express these results as equivalent SABR models. Specifically, for any given expiry date T_{ex} , we obtain the SABR model whose marginal density $Q^{SABR}(T_{ex}, X)$ is identical with the marginal density $Q^{(0)}(T_{ex}, X)$ through $O(\varepsilon^2)$.

In [2] it is shown that if $\tilde{X}$ is driven by a log normal ($\beta = 1$) SABR model,

$$(A.35a) \quad d\tilde{X} = \varepsilon \tilde{A} \tilde{X} dW,$$

$$(A.35b) \quad d\tilde{A} = \varepsilon \nu \tilde{A} dZ,$$

$$(A.35c) \quad dZ dW = \rho dT,$$

then its marginal density,

$$(A.36) \quad Q^{SABR}(T, X) \equiv Q^{SABR}(0, x, \alpha; T, X) \\ \equiv \mathbb{E} \left\{ \delta \left(\tilde{X}(T) - X \right) \middle| \tilde{X}(0) = x, \tilde{A}(0) = \alpha \right\} = \int p(0, x, \alpha; T, X, A) dA,$$

satisfies the effective forward equation

$$(A.37a) \quad Q_T^{SABR} = \frac{1}{2} \varepsilon^2 \alpha^2 e^{\varepsilon^2 \alpha \rho \nu T} \left[(1 + 2\varepsilon \rho \nu z + \varepsilon^2 \nu^2 z^2) X^2 Q^{SABR} \right]_{XX} \quad \text{for } T > 0,$$

$$(A.37b) \quad Q^{SABR}(T, X) \rightarrow \delta(X - x) \quad \text{as } T \rightarrow 0^+,$$

through $O(\varepsilon^2)$. To bring this in line with A.32a, we define the new time variable

$$(A.38a) \quad T_{new} = \frac{1 - e^{-\varepsilon^2 \Gamma T}}{\varepsilon^2 \Gamma C},$$

where C is chosen as

$$(A.38b) \quad C = \frac{1 - e^{-\varepsilon^2 \Gamma T_{ex}}}{\varepsilon^2 \Gamma T_{ex}} = e^{-\frac{1}{2} \varepsilon^2 \Gamma T_{ex}} \left\{ 1 - \frac{1}{24} \varepsilon^4 \Gamma^2 T_{ex}^2 + \dots \right\},$$

so that $T_{new}(T_{ex}) = T_{ex}$. Then the SABR model's effective forward equation becomes

$$(A.39a) \quad Q_T^{SABR} = \frac{1}{2} \varepsilon^2 C \alpha^2 e^{\varepsilon^2 (\alpha \rho \nu + \Gamma) T} \left[(1 + 2\varepsilon \rho \nu z + \varepsilon^2 \nu^2 z^2) X^2 Q^{SABR} \right]_{XX} \quad \text{for } T > 0,$$

$$(A.39b) \quad Q^{SABR}(T, X) \rightarrow \delta(X - x) \quad \text{as } T \rightarrow 0^+,$$

through $O(\varepsilon^2)$, where we are omitting the subscript “new” for brevity. Choosing

$$(A.40a) \quad \alpha^{eff} = \Delta(\alpha) e^{\frac{1}{4} \varepsilon^2 \Gamma(\alpha) T_{ex}},$$

$$(A.40b) \quad \rho^{eff} = \beta(\alpha) / \sqrt{\gamma(\alpha)},$$

$$(A.40c) \quad \nu^{eff} = \sqrt{\gamma(\alpha)},$$

makes these equations for Q^{SABR} identical to the equations A.32a, A.32b for $Q^{(0)}$, at least through $O(\varepsilon^2)$. Therefore, the SABR model's density $Q^{SABR}(T_{ex}, X)$ matches the marginal density $Q^{(0)}(T_{ex}, X)$ through $O(\varepsilon^2)$. Consequently, this SABR model accurately prices all European options with the expiry T_{ex} , at least through $O(\varepsilon^2)$, the same accuracy as the original SABR formulas.

The effective forward equation A.32a, A.32b, its coefficients A.34a-A.34f, and the SABR parameters A.40a-A.40c are our key results. These results show that European options, either standard options[5] or quadratic options[6], can be evaluated accurately by numerically solving the effective forward equation[2] or by using the implied volatility formulas contained in [5] and [6]. Implied volatility formulas for the SABR model are quoted in Appendix C.

A.5. Special cases. The main disadvantage of the above results is that the formulas rely on too many parameters. So we analyze two special cases which have a more parsimonious parameter set. These special cases only require knowing the SABR models for the individual FX rates and the correlation matrix c_{ij} between FX rates. These two cases represent opposites: in one case the volatilities are correlated as highly as possible, and in the other they are as independent as possible. Because of its theoretical importance, we also include the case where the assets are statistically identical.

A.5.1. Single stochastic vol. The volatilities of many FX rates are highly correlated; the FX markets tend to be volatile or quiet together. If we neglect idiosyncracies of individual FX volatilities, we can model all the stochastic volatilities with a single process,

$$(A.41) \quad dZ_1 = dZ_2 = \dots = dZ_n \equiv dZ.$$

This implies that

$$(A.42a) \quad \rho_{ij} = \rho_{ii} \quad \text{for all } j,$$

$$(A.42b) \quad r_{ij} = 1 \quad \text{for all } i, j.$$

For this case we define

$$(A.43a) \quad \bar{\rho} = \frac{1}{\Delta} \sum_i \rho_{ii} a_i, \quad \bar{\nu} = \frac{1}{\Delta^2} \sum_{i,j} c_{ij} a_i a_j \nu_j,$$

where

$$(A.43b) \quad \Delta^2 = \sum_{i,j} c_{ij} a_i a_j,$$

as always. In terms of these quantities the coefficients of the effective equation simplify to

$$(A.44a) \quad \beta = \bar{\rho}\bar{\nu}, \quad \gamma = \bar{\nu}^2 + \kappa,$$

$$(A.44b) \quad \Gamma = \frac{1}{\Delta^2} \sum_{i,j} c_{ij} (\nu_i - \bar{\nu}) (\nu_j - \bar{\nu}) a_i a_j - \kappa,$$

with

$$(A.44c) \quad \kappa = \frac{\bar{\rho}\bar{\nu}}{\Delta} \sum_i a_i \rho_{ii} (\nu_i - \bar{\nu}) + \frac{1}{2} \frac{\bar{\rho}^2}{\Delta^2} \sum_{i,j} c_{ij} a_i a_j (\nu_i + \nu_j - 2\bar{\nu})^2.$$

Note that $\bar{\rho}$ and $\bar{\nu}$ represent the volatility weighted correlation and vol-of-vol, while Γ and κ represent the dispersion of vols across the different FX rates. It is not uncommon for the vols ν_i to be very similar to each other, so that Γ and κ are much smaller than $\bar{\nu}^2$. In this case we can drop these terms, so that we can take $\gamma \simeq \bar{\nu}^2$ and $\Gamma \simeq 0$, at least for qualitative purposes. The effective SABR coefficients are then $\alpha^{eff} \simeq \Delta(\alpha)$, $\rho^{eff} \simeq \bar{\rho}$, $\nu^{eff} \simeq \bar{\nu}$.

A.5.2. Independent stochastic volatilities. In the opposite extreme, volatilities are as independent as possible. To match each FX rate's volatility smile, we need to allow the volatilities to be correlated with their own assets, $dW_i dZ_i = \rho_{ii} dT$. If the FX rates are correlated with each other, $dW_i dW_j = c_{ij} dT$ with $c_{ij} \neq 0$, then it is unreasonable to expect the stochastic processes to be uncorrelated with each other. Instead we assume that

$$(A.45) \quad dZ_j = \rho_{jj} dW_j + \sqrt{1 - \rho_{jj}^2} dZ_j^\perp,$$

where the Brownian motions $dZ_j^\perp$ are independent of all other stochastic processes. This implies that

$$(A.46a) \quad \rho_{ij} = c_{ij} \rho_{jj} \quad \text{for all } i, j,$$

$$(A.46b) \quad r_{ij} = c_{ij} \rho_{ii} \rho_{jj} \quad \text{for all } i, j.$$

The coefficients for the effective forward equation and the parameters for the SABR model can now be found for this case by substituting A.46a, A.46b into A.34a-A.34f.

A.5.3. Statistically identical FX rates. The last special case is one where all FX rates are modeled as being statistically identical. Then

$$(A.47a) \quad a_i \equiv a, \quad \nu_j = \nu \quad \text{for all } i,$$

$$(A.47b) \quad c_{ij} = \begin{cases} 1 & \text{for } i = j \\ c & \text{for } i \neq j \end{cases},$$

$$(A.47c) \quad r_{ij} = \begin{cases} 1 & \text{for } i = j \\ r & \text{for } i \neq j \end{cases},$$

$$(A.47d) \quad \rho_{ij} = \begin{cases} \rho_{||} & \text{for } i = j \\ \rho_{\perp} & \text{for } i \neq j \end{cases}.$$

Substituting this into A.34b-A.34f yields $\kappa = 0$, and then the coefficients reduce to

$$(A.48a) \quad \Delta^2 = a^2 n (1 - c + nc),$$

$$(A.48b) \quad \beta = \frac{\rho_{||} - \rho_{\perp} + n\rho_{\perp}}{n^{1/2} (1 - c + nc)^{1/2}} \nu,$$

$$(A.48c) \quad \gamma = \frac{1 - r + nr}{n} \nu^2,$$

$$(A.48d) \quad \Gamma = \frac{(n-1)(1-r)(1-c)}{n(1-c+nc)} \nu^2,$$

and the SABR coefficients are

$$(A.49a) \quad \alpha^{eff} = an^{1/2} (1 - c + nc)^{1/2} e^{\frac{1}{4}\varepsilon^2 \Gamma T_{ex}},$$

$$(A.49b) \quad \rho^{eff} = \frac{\rho_{||} - \rho_{\perp} + n\rho_{\perp}}{(1 - c + nc)^{1/2} (1 - r + nr)^{1/2}},$$

$$(A.49c) \quad \nu^{eff} = \frac{(1 - r + nr)^{1/2}}{n^{1/2}} \nu.$$

If, in addition, we assume that there is one stochastic vol, so $dZ_i \equiv dZ$ for all i , then $r = 1$ and $\Gamma = 0$. Since there is no difference between $\rho_{||} = \rho_{\perp} = \rho$, in this case

$$(A.50a) \quad \alpha^{eff} = an^{1/2} (1 - c + nc)^{1/2},$$

$$(A.50b) \quad \rho^{eff} = \frac{\sqrt{n}}{(1 - c + nc)^{1/2}} \rho,$$

$$(A.50c) \quad \nu^{eff} = \nu.$$

Appendix B. Cross FX SABR models.

The general cross FX model is

$$(B.1a) \quad d\tilde{X} = \varepsilon \tilde{X} \left\{ \lambda_1 \tilde{A}_1 dW_1 + \lambda_2 \tilde{A}_2 dW_2 \right\},$$

$$(B.1b) \quad d\tilde{A}_1 = \varepsilon \nu_1 \tilde{A}_1 dZ_1,$$

$$(B.1c) \quad d\tilde{A}_2 = \varepsilon \nu_2 \tilde{A}_2 dZ_2,$$

where λ_1, λ_2 are ± 1 . We first present the results for general correlation structures,

$$(B.2a) \quad dW_1 dW_2 = cdT, \quad dZ_1 dZ_2 = rdT,$$

$$(B.2b) \quad dW_i dZ_j = \rho_{ij} dT, \quad \text{for } i, j = 1, 2,$$

by specializing the results in Appendix A to two assets. We then specialize these results to the most common case, where $dZ_1 = dZ_2 \equiv dZ$.

Define

$$(B.3a) \quad \rho_{1X} \equiv \frac{a_1 \rho_{11} + a_2 \rho_{21}}{\Delta}, \quad \rho_{2X} \equiv \frac{a_1 \rho_{12} + a_2 \rho_{22}}{\Delta},$$

$$(B.3b) \quad \nu_{1X} \equiv \frac{(a_1 + ca_2) a_1}{\Delta^2} \nu_1, \quad \nu_{2X} \equiv \frac{(a_2 + ca_1) a_2}{\Delta^2} \nu_2,$$

$$(B.3c) \quad \nu_{mix} = \frac{(a_1 \rho_{11} + a_2 \rho_{21}) \nu_1 - (a_1 \rho_{12} + a_2 \rho_{22}) \nu_2}{\Delta},$$

where

$$(B.3d) \quad a_1 \equiv \lambda_1 \alpha_1, \quad a_2 \equiv \lambda_2 \alpha_2.$$

The effective forward equation A.32a, A.32b is then

$$(B.4a) \quad Q_T^{(0)} = \frac{1}{2} \varepsilon^2 \Delta^2 (\alpha) e^{\varepsilon^2 (\Gamma + \beta \Delta) T} \left[(1 + 2\varepsilon \beta z + \varepsilon^2 \gamma z^2) X^2 Q^{(0)} \right]_{XX} \quad \text{for } T > 0,$$

$$(B.4b) \quad Q^{(0)}(T, X) \rightarrow \delta(X - x) \quad \text{as } T \rightarrow 0^+,$$

where

$$(B.4c) \quad z = \frac{\ln X/x}{\varepsilon \Delta(\boldsymbol{\alpha})}.$$

Therefore, for any given expiry date T_{ex} , the SABR parameters for the equivalent log normal ($\beta = 1$) SABR model are

$$(B.5a) \quad \alpha^{eff} = \Delta(\boldsymbol{\alpha}) e^{\frac{1}{4}\varepsilon^2 \Gamma(\boldsymbol{\alpha}) T_{ex}},$$

$$(B.5b) \quad \rho^{eff} = \beta(\boldsymbol{\alpha}) / \sqrt{\gamma(\boldsymbol{\alpha})},$$

$$(B.5c) \quad \nu^{eff} = \sqrt{\gamma(\boldsymbol{\alpha})},$$

with

$$(B.6a) \quad \Delta^2(\boldsymbol{\alpha}) = a_1^2 + 2ca_1a_2 + a_2^2,$$

$$(B.6b) \quad \beta(\boldsymbol{\alpha}) = \rho_{1X}\nu_{1X} + \rho_{2X}\nu_{2X},$$

$$(B.6c) \quad \gamma(\boldsymbol{\alpha}) = \nu_{1X}^2 + 2r\nu_{1X}\nu_{2X} + \nu_{2X}^2 + \kappa(\boldsymbol{\alpha}),$$

$$(B.6d) \quad \Gamma(\boldsymbol{\alpha}) = (1 - c^2) \frac{a_1^2 a_2^2}{\Delta^4} (\nu_1^2 - 2r\nu_1\nu_2 + \nu_2^2) - \kappa(\boldsymbol{\alpha}).$$

Here,

$$(B.6e) \quad \begin{aligned} \kappa(\boldsymbol{\alpha}) = & \left\{ 3(1 - c^2) \frac{a_1 a_2}{\Delta^2} + 2c \right\} \frac{a_1 a_2}{\Delta^2} \nu_{mix}^2 \\ & - \frac{1}{\Delta} (a_2 \rho_{21} \nu_{1X} - a_1 \rho_{12} \nu_{2X}) \nu_{mix}. \end{aligned}$$

B.1. Single stochastic vol. If both volatilities are driven by the same Brownian motion, so that $dZ_1 = dZ_2 \equiv dZ$, the formulas for the SABR parameters simplify to B.5a-B.5c with

$$(B.7a) \quad \Delta^2(\boldsymbol{\alpha}) = a_1^2 + 2ca_1a_2 + a_2^2$$

$$(B.7b) \quad \beta(\boldsymbol{\alpha}) = \bar{\rho}(\boldsymbol{\alpha}) \bar{\nu}(\boldsymbol{\alpha}),$$

$$(B.7c) \quad \gamma(\boldsymbol{\alpha}) = \bar{\nu}^2(\boldsymbol{\alpha}) + \kappa(\boldsymbol{\alpha}),$$

$$(B.7d) \quad \Gamma(\boldsymbol{\alpha}) = (1 - c^2) \frac{a_1^2 a_2^2}{\Delta^4} (\nu_1 - \nu_2)^2 - \kappa(\boldsymbol{\alpha}),$$

where

$$(B.8a) \quad \bar{\rho}(\boldsymbol{\alpha}) = \frac{a_1 \rho_{11} + a_2 \rho_{22}}{\Delta},$$

$$(B.8b) \quad \bar{\nu}(\boldsymbol{\alpha}) = \frac{a_1^2 \nu_1 + ca_1 a_2 (\nu_1 + \nu_2) + a_2^2 \nu_2}{\Delta^2},$$

$$(B.8c) \quad \begin{aligned} \kappa(\boldsymbol{\alpha}) = & \left\{ 3(1 - c^2) \frac{a_1^2 a_2^2}{\Delta^4} + 2c \frac{a_1 a_2}{\Delta^2} \right\} \bar{\rho}^2 (\nu_1 - \nu_2)^2 \\ & - \frac{a_1 a_2}{\Delta^3} \{ (a_1 + ca_2) \rho_{22} \nu_1 - (a_2 + ca_1) \rho_{11} \nu_2 \} \bar{\rho} (\nu_1 - \nu_2). \end{aligned}$$

B.2. Other special cases. In the opposite extreme, the volatilities are as independent as possible. As in A.46a, A.46b the correlations are

$$(B.9a) \quad \rho_{12} = c\rho_{22}, \quad \rho_{21} = c\rho_{11},$$

$$(B.9b) \quad r_{12} = r_{21} = c\rho_{11}\rho_{22}.$$

Substituting these into B.5a-B.6e yields the coefficients for the effective forward equation and the parameters for the SABR model. As before, these new expressions are not reproduced here since there is no essential simplification..

The last special case is one where all the FX rates are modeled as being statistically identical so,

$$\begin{aligned} \text{(B.10a)} \quad & a_1 = a_2 \equiv a, \quad \nu_1 = \nu_2 \equiv \nu, \\ \text{(B.10b)} \quad & c_{12} = c_{21} \equiv c, \quad r_{12} = r_{21} \equiv r, \\ \text{(B.10c)} \quad & \rho_{11} = \rho_{22} \equiv \rho_{\parallel}, \quad \rho_{12} = \rho_{21} \equiv \rho_{\perp}. \end{aligned}$$

Substituting this into B.4c-B.6e yields $\kappa = 0$ and

$$\begin{aligned} \text{(B.11a)} \quad & \Delta^2 = 2a^2 (1 + c), \\ \text{(B.11b)} \quad & \beta = \frac{\rho_{\parallel} + \rho_{\perp}}{\sqrt{2(1 + c)}} \nu, \\ \text{(B.11c)} \quad & \gamma \equiv g = \frac{1}{2} (1 + r) \nu^2, \\ \text{(B.11d)} \quad & \Gamma \equiv \phi = \frac{1}{2} \frac{(1 - r)(1 - c)}{1 + c} \nu^2, \end{aligned}$$

so the SABR coefficients are

$$\begin{aligned} \text{(B.12a)} \quad & \alpha^{eff} = \sqrt{2(1 + c)} e^{\frac{1}{4}\varepsilon^2 \Gamma T_{ex}} a, \\ \text{(B.12b)} \quad & \rho^{eff} = \frac{\rho_{\parallel} + \rho_{\perp}}{\sqrt{(1 + c)(1 + r)}}, \\ \text{(B.12c)} \quad & \nu^{eff} = \frac{\sqrt{1 + r}}{\sqrt{2}} \nu. \end{aligned}$$

If, in addition, we assume that there is one stochastic vol, so $dZ_1 = dZ_2 \equiv dZ$, then these formulas simplify to

$$\begin{aligned} \text{(B.13a)} \quad & \alpha^{eff} = \sqrt{2} (1 + c)^{1/2} a \\ \text{(B.13b)} \quad & \rho^{eff} = \frac{\sqrt{2}\rho}{(1 + c)^{1/2}}, \\ \text{(B.13c)} \quad & \nu^{eff} = \nu. \end{aligned}$$

Appendix C. Implied volatility formulas.

C.1. Capping the stochastic vol effect. In [2] it has been shown that the reduced density $Q^{(0)}(T, X)$ for the log normal SABR model,

$$\begin{aligned} \text{(C.1a)} \quad & d\tilde{X} = \varepsilon \tilde{X} \tilde{A} dW, \\ \text{(C.1b)} \quad & d\tilde{A} = \varepsilon \nu \tilde{A} dZ, \\ \text{(C.1c)} \quad & dW dZ = \rho dT, \end{aligned}$$

satisfies the effective forward equation

$$\text{(C.2a)} \quad Q_T = \frac{1}{2} \varepsilon^2 \alpha^2 e^{\varepsilon^2 \rho \nu \alpha T} \left[(1 + 2\varepsilon \rho \nu z + \varepsilon^2 \nu^2 z^2) X^2 Q \right]_{XX}$$

through $O(\varepsilon^2)$, where

$$(C.2b) \quad z(X) = \frac{1}{\varepsilon\alpha} \log X/x,$$

and where $x = \tilde{X}(0)$ and $\alpha = \tilde{A}(0)$ are the today's FX rate and its volatility. The factor

$$(C.3) \quad 1 + 2\varepsilon\rho\nu z + \varepsilon^2\nu^2 z^2$$

encapsulates the effect of stochastic volatility. It arises because option values have a positive second derivative (gamma) with respect to the volatility $\tilde{A}$, so any changes in $\tilde{A}$ always increase the value of a vega-hedged option. Since $\tilde{A}$ is stochastic, this would lead to steady increases in the option's value. To price the option fairly, we need to increase the option's carry to balance these gains. The analysis in [2] shows that the balance requires replacing the carry $\frac{1}{2}\varepsilon^2\alpha^2 [X^2Q]_{XX}$ by

$$(C.4) \quad \frac{1}{2}\varepsilon^2\alpha^2 e^{\varepsilon^2\rho\nu\alpha T} [(1 + 2\varepsilon\rho\nu z + \varepsilon^2\nu^2 z^2) X^2Q]_{XX}.$$

The analysis presumes that this factor is not too far from one, since z is $O(1)$ and $\varepsilon \ll 1$. In practice, this factor may deviate substantially from one for very high or very low strikes, causing unduly large volatilities. To prevent this, the *stochastic vol effect* is often capped by replacing the effective forward equation by

$$(C.5a) \quad Q_T = \frac{1}{2}\varepsilon^2\alpha^2 e^{\varepsilon^2\rho\nu\alpha T} [E_c^2(\varepsilon z) X^2Q]_{XX},$$

where

$$(C.5b) \quad E_c(\varepsilon z) = \min \left\{ \sqrt{1 + 2\varepsilon\rho\nu z + \varepsilon^2\nu^2 z^2}, E_{\max} \right\}.$$

This provides an arbitrage free method for moderating the high strike and low strike vols. Typically the curb $E_{\max}$ is set between 1.5 and 3.

C.2. Explicit implied vol formulas. The capped effective forward equation for the SABR model, C.5a - C.5b, has been analyzed in [1],[2],[5]. This analysis shows that the implied (log normal) vol is

$$(C.6a) \quad \sigma_B(T_{ex}, K) = \varepsilon\alpha \cdot \frac{z(K)}{Y(\varepsilon\nu z(K))} \cdot \left\{ 1 + \varepsilon^2 \left(\frac{1}{4}\rho\nu\alpha + \frac{2-3\rho^2}{24}\nu^2 \right) T_{ex} \right\},$$

through $O(\varepsilon^2)$, where K is the strike, x and α are the current forward FX rate and its volatility, and

$$(C.6b) \quad z(K) = \frac{\log K/x}{\varepsilon\alpha}.$$

Here,

$$(C.6c) \quad Y(\varepsilon\nu z) = \frac{1}{\varepsilon\nu} \log \frac{\sqrt{1 + 2\varepsilon\rho\nu z + \varepsilon^2\nu^2 z^2} + \rho + \varepsilon\nu z}{1 + \rho} \quad \text{if } \zeta_- < z(K) < \zeta_+,$$

for moderate strikes K . For very low and very high strikes, the stochastic vol effect cap $E_{\max}$ kicks in and

$$(C.7a) \quad Y(\zeta) = Y_- - (\zeta_- - z)/E_{\max} \quad \text{if } z(K) < \zeta_-,$$

$$(C.7b) \quad Y(\zeta) = Y_+ + (z - \zeta_+)/E_{\max} \quad \text{if } z(K) > \zeta_+,$$

where

$$(C.7c) \quad \zeta_{\pm} = \frac{-\rho \pm \sqrt{E_{\max}^2 - 1 + \rho^2}}{\varepsilon \nu},$$

$$(C.7d) \quad Y_{\pm} \equiv Y(\varepsilon \nu \zeta_{\pm}) = \pm \frac{1}{\varepsilon \nu} \log \frac{E_{\max} + \sqrt{E_{\max}^2 - 1 + \rho^2}}{1 \pm \rho}.$$

The values $\zeta_{\pm}$ are where the stochastic vol factor $1 + 2\varepsilon \rho \nu z(K) + \varepsilon^2 \nu^2 z^2(K)$ first exceeds the cap $E_{\max}$.

As the strike K approaches x , the ratio $z(K)/Y(\varepsilon \nu z(K))$ approaches 1, so the at-the-money implied vol is

$$(C.8a) \quad \sigma_B^{ATM}(T_{ex}) \equiv \sigma_B(T_{ex}, x) = \varepsilon \alpha \cdot \left\{ 1 + \varepsilon^2 \left(\frac{1}{4} \rho \nu \alpha + \frac{2-3\rho^2}{24} \nu^2 \right) T_{ex} \right\},$$

and the implied volatility smile is

$$(C.8b) \quad \sigma_B(T_{ex}, K) = \sigma_B^{ATM}(T_{ex}) \frac{z(K)}{Y(\varepsilon \nu z(K))}.$$

REFERENCES

- [1] HAGAN, P.S., KUMAR, D., LESNIEWSKI, A.S., AND WOODWARD, D.E. (2002) *Managing smile risk*, Wilmott Magazine, September 2002, pp. 84 – 108.
- [2] HAGAN, P.S., KUMAR, D., LESNIEWSKI, A.S., AND WOODWARD, D.E. (2013) *Arbitrage Free SABR*, Wilmott Magazine, January (2013), pp. 1 – 16.
- [3] BALLAND, P. AND TRAN, Q. (2013). *SABR goes normal*, Risk Magazine, pp. 76-81
- [4] HAGAN, P.S., LESNIEWSKI, A.S., AND WOODWARD, D.E. (2002) *Effective medium analysis for stochastic volatility models*, Wilmott Magazine, January (2018), pp. 46 – 55.
- [5] HAGAN, P.S., KUMAR, D., LESNIEWSKI, A.S., AND WOODWARD, D.E. (2016) *Universal Smiles*, July (2016), pp. 40 – 55.
- [6] HAGAN, P.S., LESNIEWSKI, A.S., SKOUFIS, G.E, AND WOODWARD, D.E. (2019) *Explicit pricing of quadratic derivatives under SABR*, Wilmott Magazine, June 2019, pp. 40 – 52.
- [7] COLE, J.D (1968) *Perturbation Methods in Applied Mathematics* Ginn-Blaisdell.
- [8] PROTTER, M. H. AND WEINBERGER, H. F. (1984). *Maximum Principles in Differential Equations*, Springer, New York.